

Figure 5: Impact of hyper-parameters. The results of ZSL task with respect to distillation weight λ and GZSL task with respect to k for *top-k* in local head are presented.

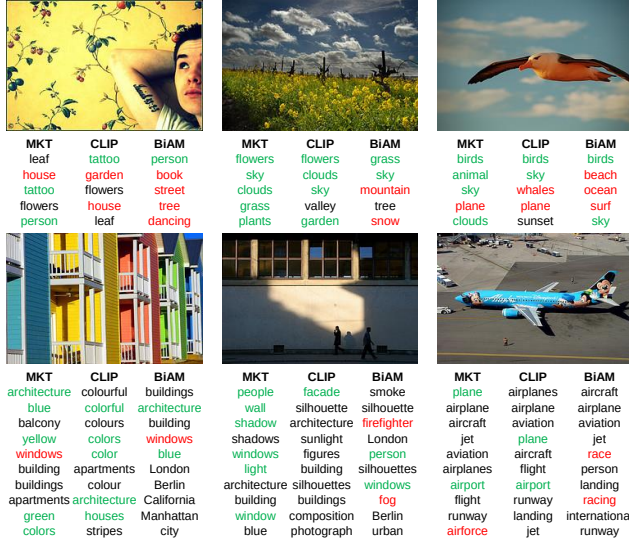


Figure 6: Comparison of predictions on test samples from NUS-WIDE. The top row shows the Top-5 prediction in ZSL task, and the bottom is the Top-10 prediction in GZSL task. True positive predictions are shown in green and the red font denotes apparently incorrect predictions.

to improve the recognition of multiple labels, so we focus more on its ability to classify all labels. Figure 5b illustrates the results of GZSL when altering k value in the local head. As k increases, F1 score reaches the highest when $k=18$. We argue that when k is too small, the local head output is sensitive to noise. On the other hand, if k is too large, the output will be less discriminative. For example, if k is set as the total patch number, *top-k* pooling will be equal to global average pooling. In contrast to F1 score, mAP tends to increase while k value increases. When k is small, the local head output tends to be discriminative but sensitive to noise, resulting in a lower mAP value. As k increases, the output becomes moderate and more resistant to noise, leading to a higher mAP value.

Qualitative Assessment

In this section, we visualize both predictions and attention maps on several samples. Figure 6 presents predictions of CLIP, BiAM and our approach on ZSL and GZSL tasks

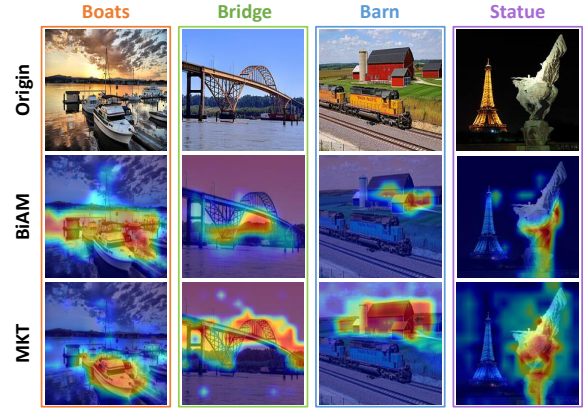


Figure 7: Comparison of Grad-CAM visualization.

respectively. Compared with CLIP, our approach produces more diverse predictions because the two-stream module captures discriminative features. Compared to BiAM, our model with VLP based label embedding can identify semantic and visual similarity among labels. For example, in the last sample of Figure 6, label “plane”, “airplane” and “aircraft” are synonymous and should have similar scores. Figure 7 illustrates the comparison of attention maps between BiAM and ours. The results show that our method can capture relevant regions more precisely. For instance, in the first column, BiAM pays attention to large irrelevant areas while our method exactly focuses on the boat region.

Conclusion

In this work, we propose an open-vocabulary based multi-modal knowledge transfer (MKT) framework for multi-label classification, which jointly exploits semantic multi-modal information in image-text pairs based VLP models. To facilitate transferring the image-text matching ability of VLP model to classification, knowledge distillation and prompt tuning are introduced. Additionally a two-stream module is proposed to capture both local and global features, leading to significant performance gains in multi-label tasks. Extensive results demonstrate that our model surpasses previous ML-ZSL methods and establishes a new state of the art for open-vocabulary multi-label classification on NUS-WIDE and Open Images datasets. This is the first work in open-vocabulary multi-label classification and it is expected to encourage future works to explore multi-modal knowledge applications in classification.

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